Automated Filter Design Tool for Impedance Matching of Wearable PVDF Ultrasound Transducers

Stephan Thurner [†], Christoph Leitner[‡], Debkalpa Goswami[†], Jörg Schröttner[†], Christian Baumgartner[†]
[†] Institute of Health Care Engineering, Graz University of Technology, Graz, Austria, [‡] Integrated Systems Laboratory, ETH Zurich, Zurich, Switzerland

Background, Motivation and Objective

Flexible piezoelectrics such as polyvinylidene fluoride (PVDF) and its co-polymeres are increasingly utilized for wearable ultrasound applications due to their conformability and cost-effective manufacturing[1]. However, transducer-transceiver impedance mismatches induce bidirectional signal reflections, degrading power transfer efficiency. Because a sensor's physical geometry inherently dictates its native impedance, dedicated matching networks are essential to minimize reflections when interfacing with standard 50Ω acquisition systems. To bridge these highly variable, application-specific architectures with standard impedance DAQs, the objective of this work was to develop an automated MATLAB tool that rapidly synthesizes matching networks for arbitrary wearable designs.

Statement of Contribution/Methods

The proposed tool integrates the Mason equivalent circuit model with transmission line theory to simulate the acoustic transducer stack. Given the transducer's impedance at resonance, the algorithm computes 50Ω matching topologies, including series, parallel, single-L, and double-L networks. It simulates the transformed impedance profile across the frequency spectrum and determines the required discrete filter components. For experimental verification, a configurable 3D-printed test environment was developed. Calculated filter networks were physically assembled and directly evaluated with PVDF samples using a vector network analyzer (VNA).

Results/Discussion

The evaluation shows good agreement between the MATLAB simulations and the VNA measurements. The tool synthesized matching networks for transducers with real impedances above, near, and below 50Ω. For a fabricated PVDF sample operating at 35 MHz, a calculated matching network improved the return loss from approximately -2 dB (unmatched) to over -30 dB (Figure 1). Deviations caused by tolerances of the discrete filter components were small relative to the simulated response. By eliminating manual trial-and-error procedures, the proposed tool enables rapid and systematic design of impedance matching networks for flexible ultrasound transducers.

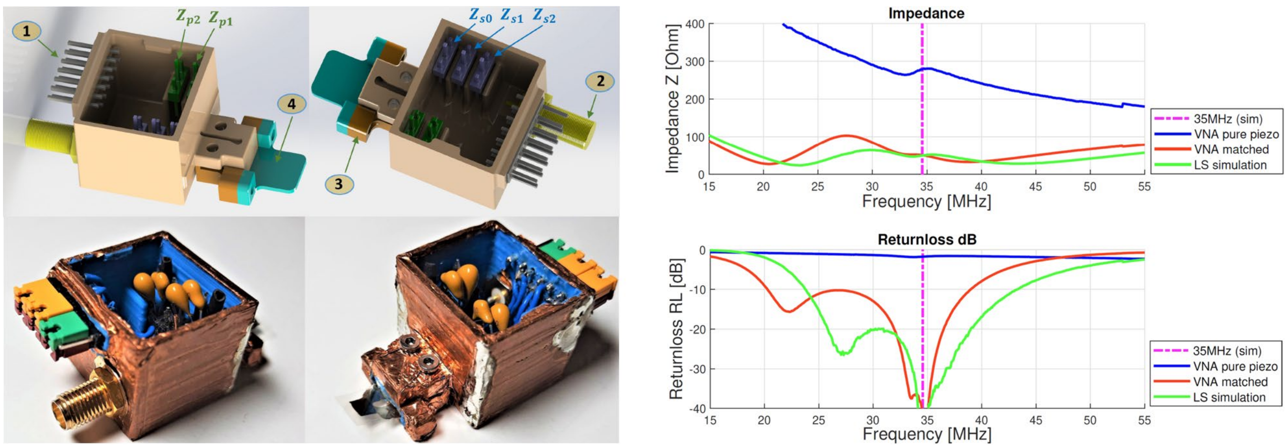

Figure 1. *Impedance matching networks for arbitrary wearable ultrasound transducer design.* ***Left:*** *Testbench with discrete filter components.* ***Right:*** *Performance comparison of unmatched (blue), simulated (green), and assembled (red) networks, detailing impedance magnitude (top) and return loss (bottom).*

[1] Saxena and Shukla, *Adv. Compos. Hybrid Mater.* 4, 8–26 (2021), doi:10.1007/s42114-021-00217-0.